Project Report

Placement Portal Application

1. Student Details

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About Me: Currently I am in Diploma level .I am interested in backend development and full-stack applications. I enjoy building structured, modular applications and improving them through clean design and database-driven logic.

2. Project Details

Placement Portal Application

Problem Statement: College need efficient systems to manage campus recruitment activities involving companies, students, and placement drives. Currently, many institutes rely on spreadsheets, emails, or manual processes, which makes it difficult to manage company approvals, track student applications, avoid duplicate registrations, and maintain placement records. A digital system is required to manage students, companies, job postings, and applications efficiently.

Approach:

The Placement Portal is implemented using Flask with a modular structure. The backend uses SQLAlchemy ORM with SQLite database. The application supports three roles:

- **Admin – manages approvals and users**
- **Company – posts jobs and manages applicants**
- **Student – applies for jobs and tracks status**

The project follows structured routing and database-driven logic with dynamic rendering using Jinja2.

3. Technologies and Frameworks Used

Technology/paragraph	Purpose
Flask	Backend web framework for routing, server logic, and request handling
SQL Alchemy	ORM used to define models and interact with the SQLite database
SQLite	Lightweight relational database for storing all application data
Jinja2	Template engine used for dynamic HTML rendering
HTML	Structure and content of the frontend pages
CSS	Styling and layout for a clean user interface
Bootstrap5	Responsive design, forms, tables, and layout components

4. Database Schema / ER Diagram

Table name	Attributes
Admin	id, name, email, password
students	id, name, email, password,phone, cgpa, branch, graduation_year,Resume,status,application relationship with student
company	id,company_name, email, password, website, hr_contact, description, address, is_approved, status

drives	id, job_title, job_description, eligibility_criteria, salary_package, location, application_deadline, status, company_id,Relationship with application
application	id, student_id, user_id, drive_id,status
	application_date, status

Relationships:

Relationship	Type
User → students	One to One
User →company	One to One
User →application	One to Many
student → Application	One to Many
company → drives	One to Many
drives → application	One to Many
student → drives (via Application)	Many to Many

5. Architecture and Features

Architecture Overview:

- Project/app.py – Main Flask application entry point, initializes the app and database, registers all route modules, and creates the default admin.

- Project/controller/database.py – Initializes SQLAlchemy and manages the database connection.

- Project/static– Resume folder and css & Img.

- /Project– Admin-specific routes for managing companies, jobs , students blacklist.

- **/templates – Jinja2 HTML templates for all pages (admin, company, student, login, registration, dashboards, profile, etc.).**

- **/static – Contains CSS files and static assets for styling the user interface.**

Implemented Features:

- **Admin, Company, and student role-based login system.**

- **student registration with profile update functionality.**

- **Admin dashboard displaying total companies, total students , total jobs , and applications.**

- **Admin controls for blacklisting, approving and companies, drives(jobs) and students.**

- **The company dashboard will display the total number of students who have applied, along with details of the students who have applied for jobs and the specific jobs they applied for.**

- **Company can update jobs status and shortlisted, selected and rejected ● Company can update jobs(drive) details which is posted by itself.**

- **Students can view Company name and job details .**

6. Video Presentation

Drive Link:

https://drive.google.com/drive/folders/1EN-Ty8k4tmwbasPR8P_wALWPstRvrDIh